# Module 9 Group Activity



## Matrix Theory

1. Consider $\mathcal{P}_2(\mathbb{R})$ with the inner product

$$\langle p, q \rangle = \frac{1}{3}\left(p(0)q(0) + p(1)q(1) + p(2)q(2)\right).$$

   (a) Show that the standard basis $\{1, t, t^2\}$ is **not** an orthogonal set with respect to this inner product.

   (b) Use the standard basis $\{1, t, t^2\}$ to find an orthonormal basis for this inner product space.

